

Maji Ndogo Data Analysis Report

Overview

This report analyzes water infrastructure, service access, safety, crime patterns, and budget allocation in Maji Ndogo using Power BI dashboards.

Scope :

- Water infrastructure
- Queue behavior
- Water quality
- Crime data
- Improvement budgets

Objective

Provide evidence for :

- Identifying water access gaps
- Measuring service efficiency
- Evaluating water safety
- Understanding crime patterns
- Guiding budget allocation

Data Sources

Data originates from the Maji Ndogo Power BI dashboards.

Includes :

- Water source data
- Population distribution
- Queue time records
- Water quality classification
- Crime records
- Budget and improvement data

Insufficient data to verify :

- Exact dataset size
- Data collection methodology
- Time range

Method

Analysis based on visual outputs :

- Map → spatial analysis
- Pie and doughnut charts → distribution
- Bar charts → comparison
- Line charts → trends over time
- Tables → detailed breakdown

No statistical recalculation performed.

Findings

Water Infrastructure

- Water sources differ across provinces and towns
- Presence of “tap in home broken” indicates system failure
- Some populations depend on shared taps or rivers

Result: Access exists in some areas but reliability varies

Queue Analysis

- Queue time changes by hour and day
- Peak periods show higher waiting times
- Queue time differs across provinces

Result: Service capacity does not match demand at all times

Water Quality

- Water classified as clean, biologically contaminated, or chemically contaminated

Result: Access does not guarantee safe water

Crime Data

- Crime types include harassment, theft, sexual assault, public intoxication, severe assault, vandalism, murder
- Distribution varies by gender
- Patterns differ by time and province

Result: Crime follows structured patterns across time and location

Budget and Improvements

- Budget distributed by improvement type
- Allocation split between rural and urban
- Number of improvements varies
- Cost per citizen differs by town

Result: Resource allocation varies across locations

Integrated Analysis

Observed relationships :

- Limited or failing infrastructure → longer queues
- High queue times → reduced access efficiency
- Unsafe water → increased health risk
- Regional differences → uneven service delivery
- Budget allocation → response to infrastructure gaps

Key Insights

- Water access is not uniform across regions
- Infrastructure failure reduces effective access
- Queue patterns reveal service pressure
- Water safety varies by source
- Crime patterns depend on time and location
- Budget allocation requires targeted planning

Limitations

- No raw dataset available for validation
- No numerical values provided for calculations
- No time-series depth beyond visual summaries

Conclusion

The dashboard shows :

- Uneven water access
- Service inefficiencies
- Safety risks
- Regional disparities
- Data supports targeted interventions and resource allocation decisions.